Activity A (ii)

Description of the proposed system

We are all on a fitness journey in one way or another. With life's hectic schedule, it's easy to lose track of where you are and where you want to be regarding your fitness. No matter whether you are trying to lose weight, put on mass, or maintain where you are, tracking fitness progress is an essential piece of your ongoing success. I have been asked by the owner of ToKa Fitness to develop a digital system that will:

- provide information about fitness training
- provide information on healthy living to help customers improve their own health and fitness
- provide access to digital content to support customers with their training
- provide access to digital content on healthy lifestyle
- encourage existing customers who have access to digital devices to use more of the services provided by ToKa Fitness.

ToKa fitness would like to help its customer track every single workout, all gains, and all food consumed in their fitness journey. These are some of the reasons mentioned during the interviews.

- People have extremely busy time schedules and it is hard for them to find slots to fit in running or other forms of workout.
- Lack of motivation is another significant reason.
- Finding people with similar goals is extremely difficult, and mostly people end up with inactive or overactive company, if found at all, which does not serve as an effective motivation booster. Not being able to track your progress is a huge roadblock in many cases.
- No proper schedule to be followed, sometimes you have the time but don't know how much and how to go about it.
- What's more, after a formidable exercise, people would get lazy or bored without any rewards or recognition. Expensive tracking and other health monitoring gadgets leave the interested people with either a big credit or without any equipment.
- Complex user interfaces and systems.
- No real data on the systems, only web fed data.
- Very little or no data analysis, as to how many calories burnt, how much time to run to improve stamina.
- No simple helpful application.

For those who log and track their progress regularly, maybe using paper or some way of recording, by using some form of electronic tracking system they would achieve some of the following aims:

- Makes it more likely to reach and surpass their goal.

- Allows them to be more efficient in time and workouts.
- Lends accountability to themselves and their goals.
- Allows for easier modifications and shows when and where changes need to be made.
- It can be motivating and reinforcing to remind them why you are doing what you are.
- Keeps you committed to their plan.

This project lends itself to a computational approach, partly because of the convenience of tracking details on the website after a workout (removing the need to carry pen and paper, and allowing users to track the details immediately, reducing the risk of errors), and partly because a computational solution will allow automatic progress tracking as it can automatically calculate and display your progress from the previous input data. This will make users' gym progress much easier and provide them with details of their progress, which would be very difficult and time-consuming to recalculate themselves every week.

The key performance indicators (KPIs) and user acceptance criteria for the proposed ToKa Fitness offers personal training sessions and advice on fitness training and healthy living to its customers and would like a website for their specific requirements. ToKa Fitness's specific requirements are that the proposed solution:

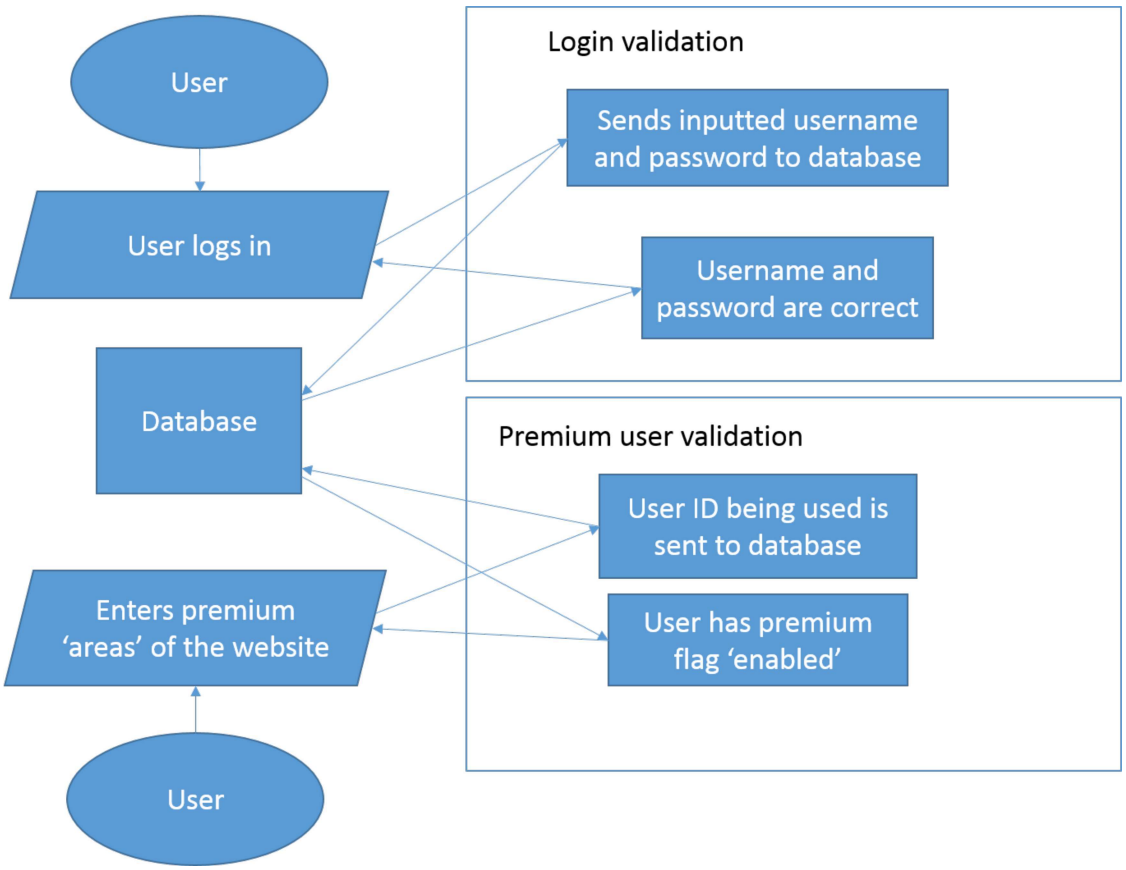
- Must have a membership system that allows the user to access extra content.
- Must allow clients to add new content without touching the code.
- Must be compatible with various devices, such as a desktop, a tablet and a smartphone.
- Should have different page layouts based on the user (e.g. client can add content, but members can only watch content).
- The layout should be simple and easy to navigate, so information should be diluted across the page (no clutter to confuse the user). Also, text information should be balanced with images and other media.
- Could use external services to provide content (e.g. use YouTube to display free videos).
- Could add advertisements to monetise the free content without making the user pay.

The system was split into sub systems that interact among each other or with the user. The interaction between the various components will be elaborated on in later sections. The system can be seen to have six components that work in unison to produce the results that help the user the best results.

- Mobile interface
- Web interface
- Database
- Location data acquisition
- Accelerometer data collection and analysis

- Profile monitoring

For the purpose of this project we will produce the first prototype, which will be the web interface.



The website can be accessed via a mobile device or a desktop computer and will have the following functionality.

Non-functional requirement is used to specify the criteria forsystem operation, which should be in contrast to functional requirement that define specific behaviours and function. In order to clarify software quality attributes, we will use the FURPS model. FURPS is an acronym for functionality, usability, reliability, performance and supportability, which is founded at Hewlett-Packard and elaborated by Grady and Caswell.

Usability is used to determine the ease of using and learning for human-made objects. First, we use PHP, JavaScript and SQL as our developing platform, which is accessible and widely used. Secondly, to make our software and website easy to use, user interface design would play a very important role. User design should make information more accessible for the user, such as a help option can be found at once when the customer encounters some problems, a traceable history for users to review their progress in previous periods in their account

Reliability is defined by the system's availability and how long it will take to recover from a failure. It is very important to our program, since our system is based on a huge amount of data. If failure occurs, it always accrues loss of data. In order to maintain high reliability, we should try our best to keep a low appearance of error in our system, and, also, if a failure is encountered, the system should have the ability to figure it out, send it to the programmer and then reset giving a reasonable explanation to t users. Performance is closely related to how fast our system can perform. As the user is training, they may want to know calories burnt

Supportability includes a variety of elements, such as maintainability, sustainability, testability and so on. As requirements above, we should make our program easier to use, not only for users, but also for people who want to modify it or add new functionality to it. In order to make sure that code can be fixed and reused easily, we should separate the whole system into several independent functions. In terms of several independent parts, people can find the main problem more quickly and rewrite the part with a problem without changing others, which will save a lot of time and manual work. Based on the above abstract information a more discrete set of requirements can be tabulated as follows.

Functional requirements:

- send email to a customer when they open a ToKa Fitness account
- user form-based authentication
- system for daily calories and movements, sleeping, fitness training and any other activates
- historical data to show trends
- transaction corrections, adjustment and cancellations
- external interface
- customer payment portal – secure considering all regulatory and legal requirements
- blog or forum for group support.

Non-functional requirements:

- performance good response time, throughput and utilisation
- scalability
- readability so should be able to view on any medium, e.g. tablet, mobile phone or desktop
- security ensure login details are secure and any important data used by the user or client

- usability.

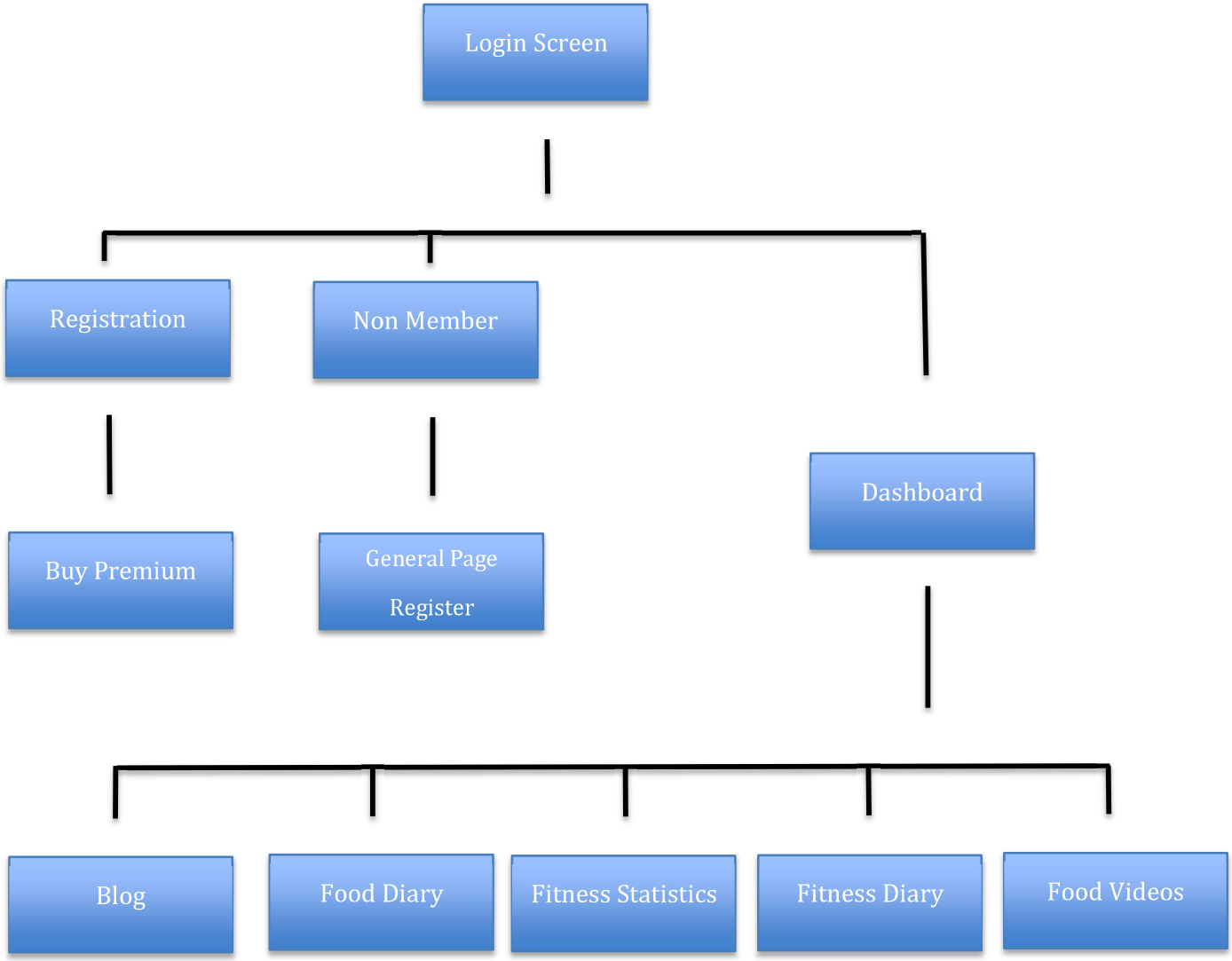
Link to:

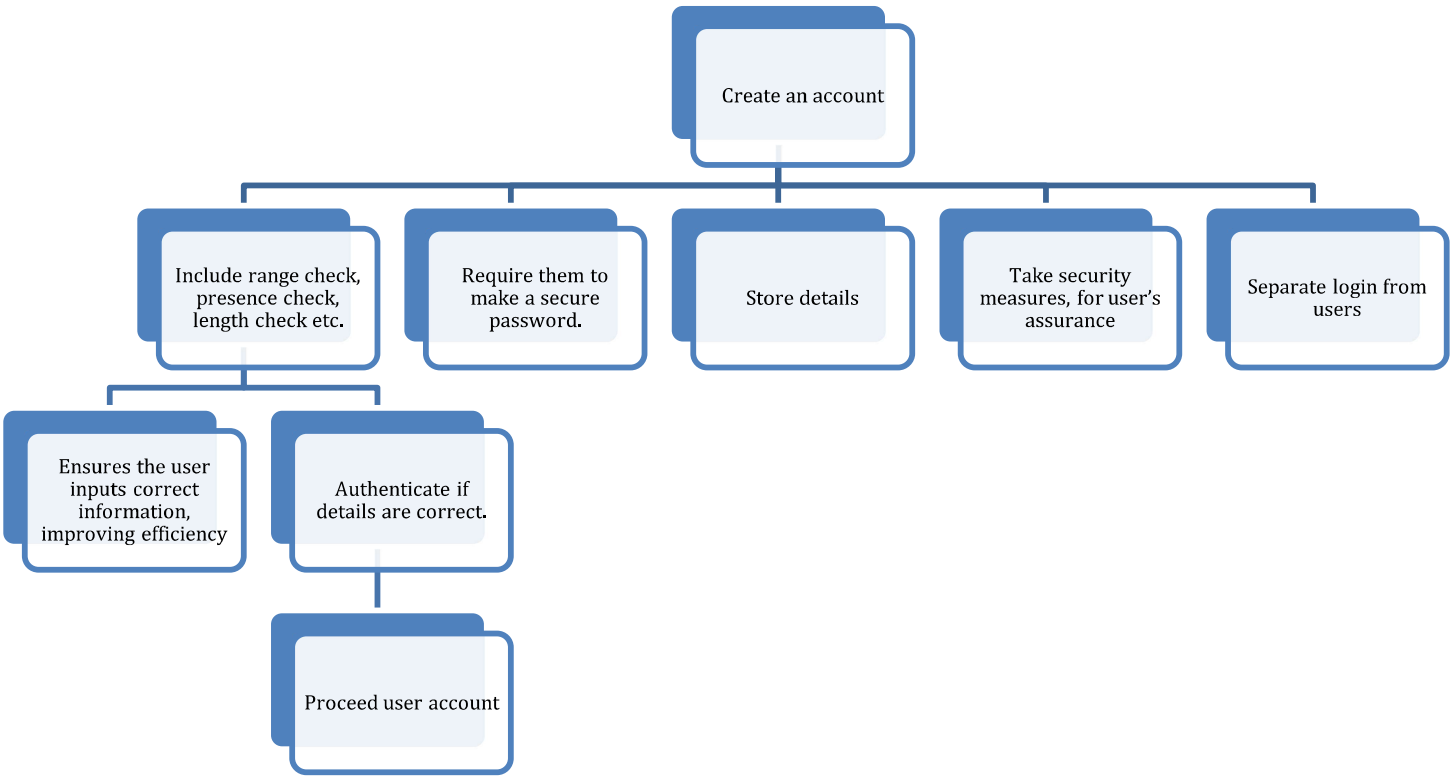
- fitness training videos
- recipe videos
- social media
- chat facility
- blog/forums
- help files on website use
- facility to record personal food notes.

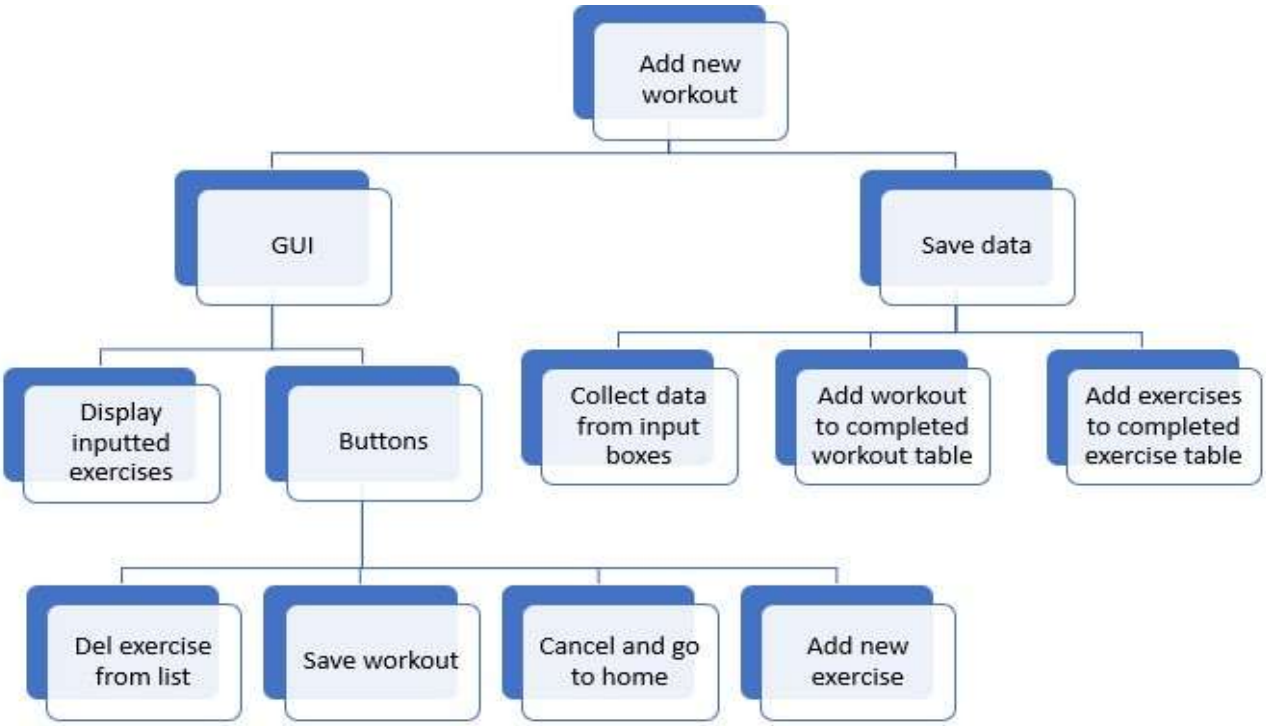
Accessibility issues to be considered:

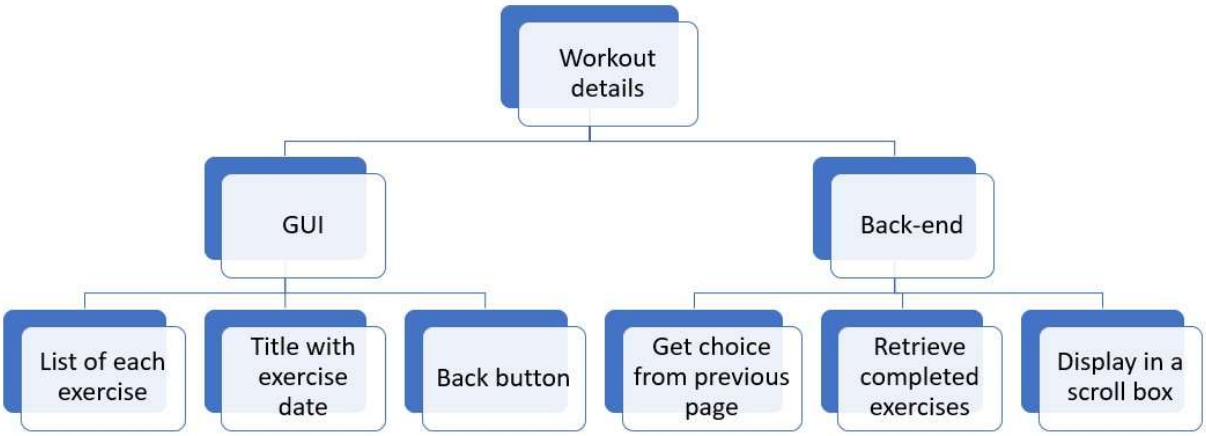
- keyboard accessibility
- colour contrast
- copyright of the use of images and videos
- colorblindness and the partially sighted user
- error messages to also have an error icon
- headings
- alternative text
- forms
- downloading materials.

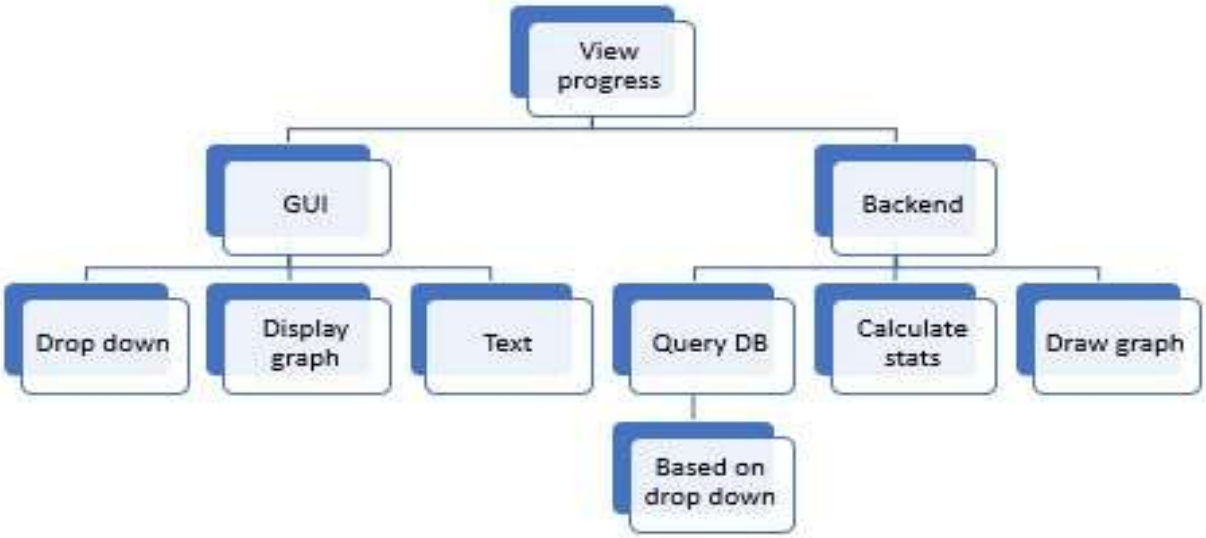
Hierarchy Diagram




Create an account
Include range check, presence check, length check etc.
Require them to make a secure password.
Store details
Take security measures, for user's assurance
Separate login from users
Ensures the user inputs correct information, improving efficiency
Authenticate if details are correct.
Proceed user account


Add new workout
GUI
Save data
Display inputted exercises
Buttons
Collect data from input boxes
Add workout to completed workout table
Add exercises to completed exercise table
Del exercise from list
Save workout
Cancel and go to home
Add new exercise





My proposed solution should provide ToKa Fitness with a professional and easy to use system that promotes their business and encourages customers to use it. All customers would have their own account. 'Free member' will have a basic level of accessibility and 'Full member' with a monthly fee

UI/UX

My stakeholders had various requests for the design of the site, including that a minimalist design would be preferred for the website, as well as indicating how many custom options they would like, and how much they can change what is visible on the main screen. There was a unanimous response that a minimalist design would be preferred over a more detailed or perhaps cluttered design. This will be implemented by taking general minimalist principles into the first prototype, such as utilising large amounts of white space with relatively small amounts of screen taken up by clickable options, a simple flat coloured background and clear labelled options.

The decision on the type of options available is less obvious, as there was no clear consensus from my stakeholders on what would be the most preferable option, as the results were quite evenly split between having all options customisable, having some hidden by default and having most hidden by default. Because of this divide I have chosen to go with the most commonly chosen option, as this will hopefully provide the best experience for users, before asking the initial question again to see if any opinions have changed. If they do not, then the other options will be used in the next prototypes in order for all solutions to be tested until a consensus is reached, otherwise a middle ground option will be used for the final release.

Another slightly vaguer set of responses to do with the UI/UX was also a focus from the stakeholders, and included the site having a fast and intuitive design, as well as providing easy access to the features that it offers. While these are harder to measure and implement, as it cannot easily be said that the site definitely offers these things, the features mentioned earlier, as well as including decisions to label fields so that they are more obvious to first time users, e.g. labelling buttons, 'backup files' or 'Enter Username', will aid in making the site more easy to use. As well as this, using commonly used icons for non-labelled options (such as using a cog to represent the settings for the site) will also help to make the site more intuitive to use. Feedback will be collected after each iteration of the prototypes in order to improve these features and create the best possible experience for the stakeholders, and eventually users.

These decisions should allow for a good initial UI and UX, but these features are liable to change as stakeholder feedback is given during the prototyping. Any suggestions for changes will be logged during the development phase, suggested to the other stakeholders and then finally implemented into the next version.

Free member	**Member**
Unregistered users should be able to: • See information about the system including guidelines and developer information. • View general information available for diet, exercise regimens and blog site. • Only view (not post on) existing forum topics. • View the community of existing users. • Sign up for an account.	Registered users should be able to: • Check in their current body measurements. • View and modify their goals. • View progress reports – daily and overall. • View BMI based on last check-in information (add in requirements). • View current day's progress. • Produce log reports. • Log out at any stage in the session.

Customers would have access to a daily fitness training planner and videos that have been verified by sports professionals and doctors. It is important that the videos and planners are safe to use and provide the correct advice and guidance.

The setup of the customised eating plans will also have to be verified by a nutritionist as they must meet the correct and current nutrition guidance. If the nutrition guidance is incorrect, the member could get very ill as the result of following the customised eating plan, and ToKa Fitness could be sued and get a bad reputation and go out of business.

Members should be able to view different workouts and monitor daily food diary statistics, depending on the customers' membership level. If the customer experience of using the website is good, they will continue to use it and promote its use to their friends and on social media.

The website will have to be secure because of members' privacy and security of user data and to comply with legal requirements such as data protection legislation and food and nutrition regulatory guidelines. Customer details must be secured by password protection and clear terms and conditions. The interface should be easy to use and navigate, be accessible across different devices and have accessibility features for users with sight loss. The password will be encrypted to prevent unauthorised access to the database.

Since there's a lot of spam and frauds, the artifact will include an email verification service that identifies any issues and possible problems. Built into the validation should be script to prevent the wrong format but this can't prevent misspelled email addresses. The service checks your mailing list for spam traps – email addresses that have been created with the intention of capturing senders who aren't following proper practices. Email verification makes sure that these addresses are removed from your mailing lists. Otherwise, if you send an email to a spam trap, you'll either be asked to remove that address from your list, or you'll be blacklisted.

For example, if someone put in this email address: john@john@dd, this is where email verification would catch it.

This project exhaustively covers many aspects of customers' requirements but there are certain limitations.

These can be summed up by the following points:

1. The ranking system is going to display the top 10 users, not all of them. However, the user can view their personal ranking.

2. Users cannot reply to others in the forum. They can only make new comments.

3. Data of calories you burned will not be calculated exactly according to the actual situation, but statistically. Time data is being collected for these parameters and they would be calculated based on algorithms.

4. Accuracy of workout is relative.

5. There is a limitation to the “push factor”, the alerts can only motivate and inform users, it is their responsibility to actually achieve targets.

6. Personal weight and body measurements must be updated by the user, the system cannot get this information.

7. User data would be available on the community section after permissions and cannot be shared otherwise. Data on the user’s personal device would never be shared directly.

User requirements

ToKa Fitness has commissioned my software development company to develop a digital system. ToKa Fitness offers personal training sessions and advice on fitness training and healthy living to its customers and would like a website for their specific requirements.

ToKa Fitness specific requirements are that the proposed solution:

- has free access with some accessibility to services
- customer section for paid content to access full services
- accessibility features for users with sight loss
- link to ‘social media’ features
- ability to view graphs and daily statistics for monitoring
- visualisation of the data to help users easily understand patterns.

The target user group is adults, male and females. Due to the legal and ethical nature of websites, it might be difficult for under 18s to follow the exercise and enter the correct details and follow the food and recipes correctly. To comply with age restrictions and guidelines, clear advice will be given to reduce health issues or injuries.

The website will be accessed via mobile devices and a computer and have compatibility across different devices, Android and iOS. It is important that the website is easy to use with on screen guidance and prompts so that the user experience is excellent in all areas of

the website to promote company image, for example using website appropriate high-quality visual assets and content. All areas of the website will comply with relevant legal regulations and guidelines.

The customer area will be secure and accessed via a password and payment made in a secure platform. This will comply with legal requirements. The privacy and security of user data is important for each member to have access to their area via a secure login.

Each customer will be able to customise workout and eating plans and generate a report on their progress for specific periods, and the level of detail of instructions in a fitness training plan will vary to allow the customer to meet their identified goal.

All links to external sources will take consideration of copyright and intellectual property and licensing requirements.

Functional Requirements			
No.	**Features**	**Priority**	**Justification**
1	Users are able to sign up and create profiles with a username and password	HIGH	This will allow many different features to work, such as the booking process, as this will allow the tutor to know who has booked the session and will allow the user to access the videos they have paid for on any device as cookies cannot transfer on all the devices the person has.
2	Users are able to log in to their own profile	HIGH	Important that the user can view their own profile details.
3	Users can view the dashboard	HIGH	This will allow the tutor to upload their timetable so that students can choose which slot they want without waiting for the tutor to confirm if they are free or not.

4	The system should contain information about the calories, weight, carbs intake, fat intake, exercise regime	HIGH	This will ensure that client will trust the organisation and make the website more reliable.
5	Users are able to pay online for membership	MEDIUM	This will allow the payment process to be much easier and less manual, which means there can be no altercation in the student not paying as it will all be recorded.

6	The system should be able to send payment confirmation to the user	MEDIUM	This will act as a receipt for the payment to the gym user and client.
7	System should have user form authentication	HIGH	This is to ensure that the user does have the email address and verifies the ID.
8	System should send a confirmation email once signed up	LOW	This will make sure that the client has all the details for signing up.
9	New system should have a blog or forum	MEDIUM	This will allow the user to raise questions and communicate with others gym members.
10	Historical data to show trends	MEDIUM	Important to generate a report that allow the gym user to view historical data and demonstrate progression.
12	External interface	HIGH	The gym user should be able to view the website on any device that allow them to see all the relevant information and perform any functionality.

Non-functional Requirements			
No.	Features	Priority	Justification
1	The system should be reliable	HIGH	This will mean that it is more trusted, which will mean more people will use the service.
2	The system should be accessible by everyone	HIGH	This will mean that anyone can access the service, which will mean that the service will be advertised to more people.
3	The system should be secure	HIGH	This will help users trust the service as they will be using their payment information on the website.
4	The system should be available at all times	HIGH	If this was not the case it would mean less people would buy the service as they cannot access it
5	PHP sessions should make the website customised to the user	MEDIUM	This will make it more customised to the user as it could be used for a faster login and it would be more personalised to the user.